

MATH 189 - Data Analysis

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1 Statistical Learning

1.1 Terminology

"Supervised Learning" means we have both predictors and responses.

"Predictors/input variables/features" are arranged in a vector.

$$X = (X_1, X_2, \dots, X_n)$$

"Output variables/responses" is a scalar where $y \in \mathbb{R}$

We think of data as coming from a model.

$$y = f(x) + \epsilon$$

$\begin{cases} f(x) \text{ is fixed but unknown function} \\ \epsilon \text{ is the random error term with mean zero, independent of } X \end{cases}$

Question: Why estimate f ?

Prediction: Given X , we want to predict y with $\hat{y} = \hat{f}(x)$

Mean square error is

$$\begin{aligned} \mathbb{E}(y - \hat{y})^2 &= \mathbb{E}(f(x) + \epsilon - \hat{f}(x))^2 \\ &= \mathbb{E}(f(x) - \hat{f}(x))^2 + 2\mathbb{E}((f(x) - \hat{f}(x))\epsilon) + \mathbb{E}(\epsilon^2) \end{aligned}$$

Examine the middle term in detail

$$2\mathbb{E}((f(x) - \hat{f}(x))\epsilon) = \mathbb{E}(f(x) - \hat{f}(x)) \cdot \mathbb{E}(\epsilon) = 0$$

We know that

$$\mathbb{E}(y - \hat{y})^2 = (f(x) - \hat{f}(x))^2 + \mathbb{E}(\epsilon^2)$$

Where the left term is reducible error and the right term is irreducible error.

"Inference" - Sometimes we want to understand the association between y and $X_1, \dots, X_p$

1. Which predictors are associated with response?
2. Is the association positive or negative?
3. Is the relationship approximately linear?

1.2 Accuracy

Training data

$$\begin{aligned} &(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n) \\ &\text{where } X_i = (X_{i1}, X_{i2}, \dots, X_{ip}) \end{aligned}$$

$n = \#$ data points (e.g. $n = 30$)

$p = \#$ predictors (e.g. $p = 3$)

Mean Square Error (MSE)

$$\text{MSE} = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{f}(x_i))^2$$

Defined on training data

$$\text{Test MSE} = \text{Ave } (y_0 - \hat{f}(x_0))^2$$

Defined on previously unseen data point (x_0, y_0)

Linear / affine model

$$f(x) = \beta_0 + \beta_1 x_1 + \dots + \beta_p x_p$$

"Fit/train the model" usually means run an optimization on your computer that uses training data and outputs the parameters $\beta_0, \beta_1, \dots, \beta_p$ that make the MSE and hopefully the test MSE small.

Classification

Data set $(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$ where $X_i = (x_{i1}, \dots, x_{ip})$ and y_i is qualitative.

Error rate is the fraction of incorrect classifications.

$$\frac{1}{n} \sum_{i=1}^n \mathbb{I}\{y_i \neq \hat{y}_i\}$$

$$\mathbb{I}\{y_i \neq \hat{y}_i\} = \begin{cases} 1, & y_i \neq \hat{y}_i \\ 0, & y_i = \hat{y}_i \end{cases}$$

Test error rate

$$\text{Ave}(\mathbb{I}\{y_0 \neq \hat{y}_0\})$$

Where (x_0, y_0) is a previously unseen test data point.

K-nearest neighbors (KNN)

The KNN classifier is defined by letting $\eta_0 =$ set of indices $i \in \{1, \dots, n\}$ for K nearest neighbors of X_0 among the input data.

$\hat{y}_0 = j$ where j maximizes

$$\sum_{i \in \eta_0} \mathbb{I}\{y_i = j\}$$

which is the number of nearest neighbors (out of K) assigned label j .

2 Linear Regression

2.1 Simple Linear Regression

Underlying statistical model: $y = \beta_0 + \beta_1 X + \epsilon$ where y is the quantitative response and a single predictor X which is why it is called simple.

Prediction model: $\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 X$

$\hat{\beta}_0, \hat{\beta}_1$ are estimates based on training data, where hat means estimated quantity.

Estimating coefficients

The residual sum of squares (RSS)

$$\begin{aligned} \text{RSS} &= \sum_{i=1}^n (y_i - \hat{y}_i)^2 \\ &= \sum_{i=1}^n (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i)^2 \end{aligned}$$

Choose $\hat{\beta}_0, \hat{\beta}_1$ to minimize RSS.

$$\begin{aligned} \frac{\partial \text{RSS}}{\partial \hat{\beta}_0} &= \frac{\partial}{\partial \hat{\beta}_0} \sum_{i=1}^n (y_i - \hat{\beta}_0 - \hat{\beta}_1 X)^2 \\ &= 2 \sum_{i=1}^n (\hat{\beta}_0 + \hat{\beta}_1 X_i - Y_i) \\ \implies \text{RSS is } &\begin{cases} \text{decreasing for really small } \hat{\beta}_0 \\ \text{increasing for really big } \hat{\beta}_0 \end{cases} \end{aligned}$$

RSS is optimal (in $\hat{\beta}_0$) when

$$\begin{aligned}
0 &= \frac{1}{n} \sum_{i=1}^n (\hat{\beta}_0 + \hat{\beta}_1 X_i - Y_i) \\
&= \hat{\beta}_0 + \hat{\beta}_1 \left(\frac{1}{n} \sum_{i=1}^n X_i \right) - \left(\frac{1}{n} \sum_{i=1}^n Y_i \right) \\
&= \hat{\beta}_0 + \hat{\beta}_1 \bar{x} - \bar{y} \\
\hat{\beta}_0 &= \bar{y} - \hat{\beta}_1 \bar{x}
\end{aligned}$$

Where $\begin{cases} \bar{x} = \frac{1}{n} \sum_{i=1}^n X_i \text{ is the mean predictor} \\ \bar{y} = \frac{1}{n} \sum_{i=1}^n Y_i \text{ is the mean response} \end{cases}$

Covariance

$$\begin{aligned}
\text{Cov}(X_i, Y_i)_{i=1}^n &= \frac{1}{n} \sum_{i=1}^n X_i Y_i - \bar{X} \bar{Y} \\
&= \frac{1}{n} \sum_{i=1}^n (X_i - \bar{x})(Y_i - \bar{y})
\end{aligned}$$

Variance

$$\begin{aligned}
\text{Var}(X_i)_{i=1}^n &= \text{Cov}(X_i, X_i)_{i=1}^n \\
&= \frac{1}{n} \sum_{i=1}^n (X_i - \bar{x})^2
\end{aligned}$$

From this we can get our optimal $\hat{\beta}_1$

$$\begin{aligned}
\hat{\beta}_1 &= \frac{\text{Cov}(X_i, Y_i)_{i=1}^n}{\text{Var}(X_i)_{i=1}^n} \\
&= \frac{\frac{1}{n} \sum_{i=1}^n (X_i - \bar{x})(Y_i - \bar{y})}{\frac{1}{n} \sum_{i=1}^n (X_i - \bar{x})^2}
\end{aligned}$$

2.2 Goodness of Fit & Correlation

Assume an underlying statistical model

$$y = \beta_0 + \beta_1 X + \epsilon$$

We use training data $(X_i, Y_i)_{i=1}^n$ to generate prediction model $\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 X$
The formulas for $\hat{\beta}_0$ and $\hat{\beta}_1$ to minimize RSS:

$$\text{RSS} = \sum_{i=1}^n (y_i - \hat{y}_i)^2 \begin{cases} \hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{X} \\ \hat{\beta}_1 = \frac{\text{Cov}(X_i, Y_i)_{i=1}^n}{\text{Var}(X_i)_{i=1}^n} \end{cases}$$

Goodness of fit

$$\begin{aligned} \text{RSS} &= \sum_{i=1}^n (y_i - \hat{y}_i)^2 \\ &= \sum_{i=1}^n (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i)^2 \\ &= \sum_{i=1}^n \left((y_i - \bar{y}) - \hat{\beta}_1 (x_i - \bar{x}) \right)^2 \\ &= \sum_{i=1}^n (y_i - \bar{y})^2 - 2\hat{\beta}_1 \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y}) + \hat{\beta}_1^2 \sum_{i=1}^n (x_i - \bar{x})^2 \\ &= \sum_{i=1}^n (y_i - \bar{y})^2 - \frac{\left[\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y}) \right]^2}{\sum_{i=1}^n (x_i - \bar{x})^2} \end{aligned}$$

RSS scales with the number of data points n and also scales with y^2 units.
→ We can fix the scaling and get out a goodness-of-fit score (R^2) between 0 and 1.

$$R^2 = 1 - \frac{\text{RSS}}{\text{TSS}}$$

$$\text{RSS} = \sum_{i=1}^n (y_i - \hat{y}_i)^2 \qquad \text{TSS} = \sum_{i=1}^n (y_i - \bar{y})^2$$

So for simple linear regression,

$$\begin{aligned}
 R^2 &= 1 - \frac{\text{RSS}}{\text{TSS}} \\
 &= \frac{\text{TSS} - \text{RSS}}{\text{TSS}} \\
 &= \frac{\left[\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y}) \right]^2}{\sum_{i=1}^n (x_i - \bar{x})^2 \sum_{i=1}^n (y_i - \bar{y})^2} \\
 R &= \frac{\text{Cov}((x_i), (y_i))}{\text{Var}(x_i)^{1/2} \text{Var}(y_i)^{1/2}}
 \end{aligned}$$

Long story short R^2 is the fraction of variance explained by the model.

`ex` $\hat{\text{sales}} = 7.0 + 0.048 \cdot \text{TV}$
 $R^2 = 0.61 = 61\%$ $r \approx 0.78$ pretty strong positive correlation

2.3 Regression Table

`ex`

	Coefficient	Std. error	t-statistic	p-value
Intercept (β_0)	7.0	0.46	15	<0.0001
TV (β_1)	0.048	0.0027	18	<0.0001

Std error is the error bar on linear coefficients.

We know that
$$\begin{cases} \beta_0 \approx \hat{\beta}_0 \pm \text{SE}(\hat{\beta}_0) \\ \beta_1 \approx \hat{\beta}_1 \pm \text{SE}(\hat{\beta}_1) \end{cases}$$

Assume the underlying statistical model is true!

$$y_i = \beta_0 + \beta_1 X_i + \epsilon_i$$

where ϵ_i are independent $\eta(0, \sigma^2)$ random variables

If we generate N data points (x_i, y_i) from the model, then

$$\begin{cases} \text{Var}(\hat{\beta}_0)^{1/2} \approx \text{Std. error}(\hat{\beta}_0) \\ \text{Var}(\hat{\beta}_1)^{1/2} \approx \text{Std. error}(\hat{\beta}_1) \\ \mathbb{E}(\hat{\beta}_0) = \beta_0 \\ \mathbb{E}(\hat{\beta}_1) = \beta_1 \end{cases}$$

2.4 Multiple Linear Regression

The underlying statistical model that we assume is true.

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_p x_p$$

where y is response, β_0 is the intercept, $\beta_1 x_1, \dots, \beta_p x_p$ are predictors, and ϵ is the mean zero noise.

$$\boxed{ex} \quad \text{sales} = \beta_0 + \beta_1 \text{TV} + \beta_2 \text{Radio} + \beta_3 \text{Newspaper} + \epsilon$$

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x_1 + \dots + \hat{\beta}_p x_p$$

We want to find the best $\hat{\beta}$ to minimize RSS, where $\hat{\beta}$ are estimated quantities.

$$\begin{aligned} RSS &= \sum_{i=1}^n (y_i - \hat{y}_i)^2 \\ &= \sum_{i=1}^n (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_{i1} - \dots - \hat{\beta}_p x_{ip})^2 \\ &= \left\| \begin{bmatrix} y_1 - \hat{\beta}_0 - \hat{\beta}_1 x_{11} - \dots - \hat{\beta}_p x_{1p} \\ \vdots \\ y_n - \hat{\beta}_0 - \hat{\beta}_1 x_{n1} - \dots - \hat{\beta}_p x_{np} \end{bmatrix} \right\|^2 \\ &= \|Y - X\hat{\beta}\|^2 \end{aligned}$$

$$\text{where } y = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix} \quad \hat{\beta} = \begin{bmatrix} \hat{\beta}_0 \\ \hat{\beta}_1 \\ \vdots \\ \hat{\beta}_p \end{bmatrix} \quad X = \begin{bmatrix} 1 & x_{11} & \dots & x_{1p} \\ 1 & x_{21} & \dots & x_{1p} \\ & & \ddots & \\ 1 & x_{n1} & \dots & x_{np} \end{bmatrix}$$

The best $\hat{\beta}$ to minimize Euclidean norm² for $\|Y - X\hat{\beta}\|^2$ is $\hat{\beta} = X^+ Y = (X^T X)^{-1} X^T Y$

	Coefficient	Std. error	t-statistic	p-value
Intercept	$\hat{\beta}_0$ 2.939	0.3119	9.42	<0.0001
TV	$\hat{\beta}_1$ 0.046	0.0014	32.81	<0.0001
Radio	$\hat{\beta}_2$ 0.189	0.0086	32.89	<0.0001
Newspaper	$\hat{\beta}_3$ -0.001	0.0059	-0.18	0.8599

Note: the standard error is NOT $Var(\hat{\beta}_i)^2$ but rather $\hat{Var}(\hat{\beta}_i)^2$

Hypothesis testing

State a hypothesis, test it with data (testing the likelihood such data could possibly be generated if the hypothesis were true).

H_0 - the null hypothesis: $y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + \epsilon$ for some coefficients $\beta_0, \beta_2, \beta_3$ with $\beta_1 = 0$ with $\epsilon \sim N(0, \sigma^2)$ for some variance $\sigma^2 \geq 0$. Also, observations are independent.

H_1 - the alternative hypothesis: $y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + \epsilon$ where everything above holds except $\beta_1 \neq 0$

Assume H_0 . Then

$$\frac{\hat{\beta}_1}{\hat{SE}(\hat{\beta}_1)} = t \sim T_{n-p-1}$$

2.5 Generative Models for Classification

Predictors

X or $X_1, X_2, \dots, X_n$

Response

$y \in \{0, 1\}$ or $y \in \{1, 2, \dots, k\}$

Review

Simple logistic regression

$$P = \text{prob}\{y = 1\} = \frac{\exp(\beta_0 + \beta_1 X)}{1 + \exp(\beta_0 + \beta_1 X)}$$

Note: This doesn't make any assumptions about how predictors are distributed.

For each $k = 1, 2, \dots, K$:

Our statistical model ('mixture of Gaussians') says $y = k$ with probability π_k ($\sum_{k=1}^K \pi_k = 1$)

Given $y = k$, $X \sim N(\mu_k, \Sigma_k)$ where Σ is the covariance matrix symbol. Here is the general vector-valued format:

$$X = (X_1, \dots, X_p) \quad \Sigma_k = \begin{bmatrix} & \\ & \end{bmatrix} \text{ pxp matrix}$$

Where μ_k has Kp parameters and Σ_k has $K \frac{p(p+1)}{2}$ parameters
When $p = 1$, we can write

$$X \sim N(\mu_k, \sigma_k^2)$$

In approaches like Linear Discriminant Analysis (LDA), Quadratic Discriminant Analysis (QDA), and Naive Bayes (NB) use simple estimators for Π_k, μ_k, Σ_k that we can calculate fast.

ex Linear discriminant analysis (LDA) make assumptions of common covariance structure $\Sigma = \Sigma_1 = \Sigma_2 = \dots = \Sigma_k$ which reduces a lot of complexity. Given observations $(X_i, y_i)_{1 \leq i \leq n}$, we can estimate for $k = 1, \dots, K$

$$\begin{aligned} \hat{\Sigma} &= \frac{n_k}{n}, \text{ where} \\ n &= \#\{i : y_i = k\} \\ &= \text{number of observations in class } k \\ \hat{\mu}_k &= \frac{1}{n_k} \sum_{i: y_i = k} X_i \end{aligned}$$

Last estimate common covariance by

$$\begin{aligned} \hat{\Sigma} &= \frac{1}{n - K} \sum_{k=1}^K \sum_{i: y_i = k} (X_i - \mu_k)(X_i - \mu_k)^T \\ \mathbb{E} \hat{\Sigma} &= \Sigma \text{ if the model is true} \end{aligned}$$

ex Quadratic discriminant analysis is pretty much the same except DIFFERENT covariances $\Sigma_1, \dots, \Sigma_k$ for each class.

$$\begin{aligned} \implies \hat{\Pi}_k &= \frac{n_k}{n}, \hat{\mu}_k = \frac{1}{n_k} \sum_{i: y_i = k} X_i \\ \implies \hat{\Sigma}_k &= \frac{1}{n_k - 1} \sum_{i: y_i = k} (X_i - \mu_k)(X_i - \mu_k)^T \\ \implies &K \frac{p(p+1)}{2} \text{ covariance free parameters.} \end{aligned}$$

ex Naive Bayes is pretty much the same except DIFFERENT covariances $\Sigma_1, \dots, \Sigma_k$ that are also diagonal matrices.
 $\implies Kp$ covariance free parameters with the same estimators as QDA.

$$\text{diag} \left(\sum_k^n \right) = \frac{1}{n_k - 1} \sum_{i: y_i = k} (X_i - \mu_k) \odot (X_i - \mu_k)$$

Suppose you have the statistical model with probability Π_k , we set $y = k$ and draw $X \sim N(\mu_k, \Sigma_k)$. Suppose we have estimated the parameters well - see a new predictor vector $X = (X_1, \dots, X_p)$ - How do you determine class probabilities $P\{y = k|X\}$ and make predictions?

Prediction part is easy - just predict based on the highest probability class:
 $\hat{y} = k$ for $k = \text{argmax}_{1 \leq j \leq K} P\{y = j|X\}$

So what is $P\{y = k|X\}$? Bayes' rule tells us

$$\begin{aligned} P\{y = k|X\} &= \frac{P\{x|y = k\} P\{y = k\}}{\sum_{j=1}^K P\{X|y = j\} P\{y = j\}} \\ &= P(X) \\ &= \frac{f_k(X) \Pi_k}{\sum_{j=1}^K f_j(X) \Pi_j} \end{aligned}$$

where Gaussian density $\Pi_j(X) = \frac{1}{(2\pi)^{p/2} (\det \Sigma)^{1/2}} \exp\left(\frac{-1}{2} (X - \mu_j) \Sigma_j (X - \mu_j)\right)$